

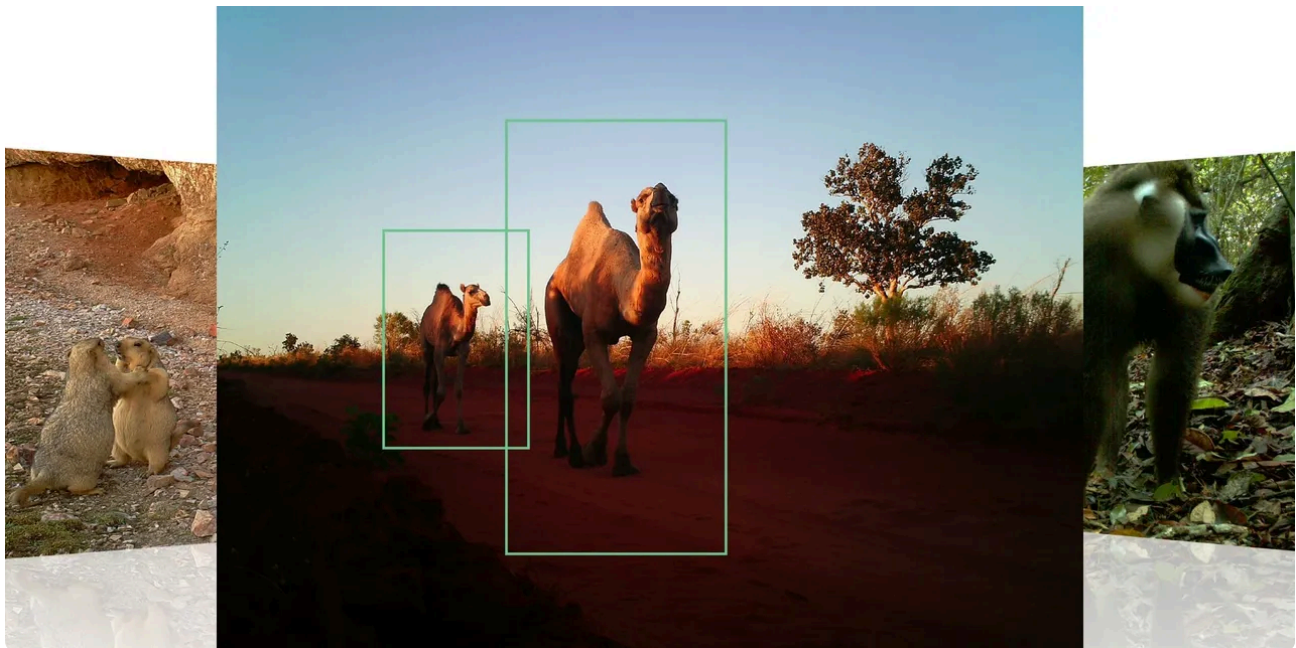
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Using the power of AI to identify and track species

Now open source, the SpeciesNet model saves time and saves wildlife

By [Abby Hehmeyer](#)

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In the world of wildlife conservation, every image tells a story. But imagine having millions of wildlife images that you had to sort through by hand – this slow and tedious process could take months. Now imagine if you could analyze those millions of wildlife images in just a few minutes. Artificial

Skip to main content e ability to do this heavy lifting, and with a free, open-access tool like SpeciesNet, anyone can now use the power of AI to sort and classify their camera photos automatically.

Every year, WWF staff deploy thousands of motion-triggered cameras, or “camera traps”, to monitor wildlife populations. These cameras generate millions of images, but manually sorting and analyzing this wealth of information is a slow and labor-intensive process. That’s why WWF uses **Wildlife Insights**, a revolutionary platform developed in collaboration with other conservation organizations and Google, which uses the power of artificial intelligence (AI) and machine learning to automatically identify animals in images. The Wildlife Insights AI model, called SpeciesNet, is also now **freely available** online for anyone to use and improve on, whether you’re a researcher, conservationist, or wildlife enthusiast.

The SpeciesNet model has been part of the Wildlife Insights platform since 2019 and has been trained on a dataset of over 65 million images contributed by WWF and other conservationists. As a result, it can recognize a wide range of species with remarkable accuracy. It detects 99.4% of images containing animals and, when the model predicts an animal is present, it is correct 98.7% of the time. Additionally, the model is accurate 94.5% of the time when it makes a species-level prediction.

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Camera trap image of a jaguar named "Chio," identified in Tahuamanu timber concession
© WWF Peru

Releasing SpeciesNet as an open-source tool accelerates advancements in wildlife monitoring, because anyone can access the model, use it in their projects, and even contribute to improving it. Whether you're monitoring wildlife in the Amazon, the Arctic, or your own backyard, SpeciesNet is a powerful tool to support biodiversity research and conservation efforts. Developers across the conservation community can now help to further refine the model to identify more species, more accurately.

How has SpeciesNet been used in wildlife conservation?

SpeciesNet represents a significant step forward in harnessing technology to protect wildlife. Using AI to automatically identify and catalog camera trap images dramatically accelerates research, enhances data sharing, and empowers global conservation efforts.

[Skip to main content](#) Jaguar Conservation in Peru

A biodiversity hotspot, the Amazon rainforest in Peru is home to an extraordinary array of wildlife, including the elusive jaguar (*Panthera onca*). Unfortunately, deforestation, habitat fragmentation, and human encroachment pose significant threats to jaguar populations and other wildlife. To combat these threats, WWF and partners use camera traps to better understand jaguar densities and the relative abundance of other species. The images are then processed using the SpeciesNet AI model in Wildlife Insights.

In Peru's Tahuamanu region, 136 camera traps are deployed across timber and conservation forest concessions to better understand how wildlife utilize these human-use areas. Preliminary results indicate the presence of 37 individual jaguars, as well as other species such as the South American tapir, peccary (a pig-like ungulate), and ocelot. These data help us assess the conservation status of these species and evaluate the effectiveness of our interventions, such as sustainable forest management, regenerative cattle ranching with a zero-deforestation approach, and other initiatives developed in partnership with local communities.



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Tracking Sky-high Species

In Tahuamanu province, department of Madre de Dios, Perú, the roads through timber and Brazilian nut forest concessions can be a hazard for arboreal wildlife, making it challenging for them to travel between tree covered areas. In response, WWF installed innovative canopy bridges over the roads to help above-ground species cross safely and cameras were installed to evaluate whether they were being used. Data processed using the SpeciesNet model confirmed six species, including five primates and one carnivore, the kinkajou (*Potos flavus*), used the canopy bridges, confirming these bridges facilitate greater movement and safety for arboreal wildlife.



A collared peccary (*Dicotyles tajacu*) in Tahuamanu timber concession.

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A South American tapir (*Tapirus terrestris*), also in the Tahuamanu timber concession.

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How to use SpeciesNet

For those who need a platform that easily and quickly runs SpeciesNet, the model remains available within Wildlife Insights, a global resource for biodiversity monitoring and management. Wildlife Insights will continue to provide the latest version of the model as it is updated.

For those practitioners and developers who want to use SpeciesNet on its own, the [GitHub repository](#) provides access to the model, documentation, and resources necessary for running and adapting the model. We invite researchers, conservationists, and citizen scientists to contribute to the project, refine the model, expand its capabilities, and join us in advancing the field of wildlife conservation through technology.

Learn more

[Wildlife conservation](#)[Artificial Intelligence](#)[Jaguar](#)

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How you can help

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Support Tribes restore buffalo to the Great Plains

The Indian Buffalo Management Act would strengthen support for Tribal Nations who are working to bring buffalo back from the brink of extinction.

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